

Для начала необходимо сформировать дефолтный конфиг.

```
wild4est | wild4est-laptop | ~/Desktop/MyFavoriteFolder/learn/Eltex/LinuxKernel/linux-7.0.11 $ make ARCH=arm defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
*** Default configuration is based on 'multi_v7_defconfig'
#
# configuration written to .config
#
```

Рисунок 1 — формирование дефолтного конфига

Затем нужно собрать образ ядра.

```
wild4est | wild4est-laptop | ~/Desktop/MyFavoriteFolder/learn/Eltex/LinuxKernel/linux-7.0.11 $ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- -j8 zImage
SYNC include/config/auto.conf.cmd
* Restart config...
*
* Kernel Features
*
```

Рисунок 2 — сборка zImage

После собрать файлы dtb.

```
wild4est | wild4est-laptop | ~/Desktop/MyFavoriteFolder/learn/Eltex/LinuxKernel/linux-7.0.11 $ make ARCH=arm -j8 dtbs
SYNC include/config/auto.conf.cmd
DTC arch/arm/boot/dts/actions/owl-s500-cubieboard6.dtb
DTC arch/arm/boot/dts/airoha/en7523-evb.dtb
DTC arch/arm/boot/dts/amazon/alpine-db.dtb
DTC arch/arm/boot/dts/arm/vexpress-v2p-ca9.dtb
DTC arch/arm/boot/dts/amlogic/meson8-minix-neo-x8.dtb
DTC arch/arm/boot/dts/aspeed/aspeed-ast2500-evb.dtb
DTC arch/arm/boot/dts/axis/artpec6-devboard.dtb
DTC arch/arm/boot/dts/allwinner/sun4i-a10-a1000.dtb
```

Рисунок 3 — сборка файлов dtb

Теперь нужно разместить файл zImage и dtb файл в отдельной директории.

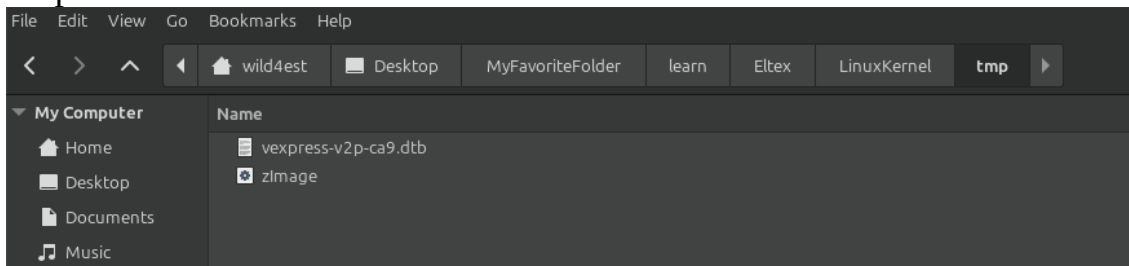


Рисунок 4 — директория с zImage файлом и dtb файлом

Затем из текущей папки можно запустить полученный образ через qemu.

```
wild4est | wild4est-laptop | ~/Desktop/MyFavoriteFolder/learn/Eltex/LinuxKernel/tmp $ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nogr
aphic
[ 0.000000] Booting Linux on physical CPU 0x0
[ 0.000000] Linux version 7.0.11 (wild4est@wild4est-laptop) (arm-linux-gnueabi-gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #1 SMP Sun Jun 7 14:26:02 +07 2026
[ 0.000000] CPU: ARMv7 Processor [410fc090] revision 0 (ARMv7), cr=10c5387d
[ 0.000000] CPU: PIPT / VPIPT nonaliasing data cache, VPIPT nonaliasing instruction cache
[ 0.000000] OF: fdt: Machine model: V2P-CA9
[ 0.000000] Memory policy: Data cache writeback
[ 0.000000] efi: UEFI not found.
[ 0.000000] cma: Failed to reserve 64 MiB
[ 0.000000] Reserved memory: created DMA memory pool at 0x4c000000, size 8 MiB
[ 0.000000] OF: reserved mem: initialized node vram@4c000000, compatible id shared-dma-pool
[ 0.000000] OF: reserved mem: 0x4c000000..0x4c7fffff (8192 KiB) nomap non-reusable vram@4c000000
[ 0.000000] CPU: All CPU(s) started in SVC mode.
[ 0.000000] Zone ranges:
[ 0.000000] DMA [mem 0x0000000000000000-0x0000000067ffffff]
[ 0.000000] Normal empty
[ 0.000000] HighMem empty
```

Рисунок 5 — запуск образа

По итогу проделанной работы получаем ошибку Kernel panic из-за отсутствия файловой системы.

```
[ 2.160004]
[ 2.160911] Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0)
[ 2.171638] CPU: 0 UID: 0 PID: 1 Comm: swapper/0 Not tainted 7.0.11 #1 VOLUNTARY
[ 2.172222] Hardware name: ARM-Versatile Express
[ 2.172691] Call trace:
[ 2.173417] unwind_backtrace from show_stack+0x10/0x14
[ 2.174733] show_stack from dump_stack_lvl+0x54/0x68
[ 2.175082] dump_stack_lvl from vpanic+0x234/0x454
[ 2.175390] vpanic from do_panic_on_target_cpu+0x0/0x14
[ 2.176479] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0) ]---
```

Рисунок 6 — ошибка Kernel panic